Rosseland Mean Opacity for Expansion Opacity from Bound-Bound Lines

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Abstract

We present a new algorithm for calculating Rosseland mean opacity that accounts for expansion opacity effects from bound-bound atomic transitions. The method smooths the frequency-dependent opacity using a $\tau = 1$ window, where for each frequency the distance required to reach optical depth unity is calculated. The smoothed opacity is then determined from $\kappa = 1/(\rho\Delta x)$, where Δx is the path length for $\tau = 1$. Monte Carlo radiative transfer simulations using this new Rosseland opacity yield results identical to full frequency-group calculations, demonstrating that the averaging successfully captures the essential physics of expansion opacity.

Keywords:
opacity, radiative transfer, Rosseland mean, expansion opacity, bound-bound transitions

1. Introduction

The interaction between radiation and matter is primarily determined by the material properties such as composition, density, and temperature, as well as the spectral characteristics of the radiation. In expanding plasmas, due to the Doppler effect, as photons travel through the plasma, their frequency changes. This effect is mainly important for bound-bound transitions, due to their high gradient in frequency which cause photons to interact with different lines as they travel through the plasma. This effect is known as expansion opacity, and it is important for many astrophysical applications, such as Type Ia supernovae and neutron star mergers. Since in most applications, the velocity of the plasma is no more the few percents of the speed of light, the frequency shift due to the Doppler shift is approximately $\Delta\nu = \nu_0 \frac{\Delta v}{c}$.

For non-moving plasmas, the radiation flux can be usually approximated well using the Rosseland mean opacity, which is a weighted average of the opacity over all frequencies. However, for expanding plasmas, the Rosseland mean opacity is not a good approximation, since the opacity changes as the photon travels through the plasma. And therefore the average distance a photon travels depends on his direction of movement and the plasma's velocity gradient.

In this paper we present a new algorithm for calculating Rosseland mean opacity that properly accounts for expansion opacity effects from bound-bound lines. The method assume a photon which travels "outwards" from the center of the expanding plasma and assumes bolometric expansion ($r = v \cdot t$) and thus gives the opacity as a function of time in addition to the density and temperature $\kappa(T, \rho, t)$.

The structure of the paper is as follows: In Section 2 we describe the expansion opacity and how it is calculated. In Section XXX, we discuss the effective Rosseland mean opacity calculation for expansion opacity. In Section 3 we describe the results given by the algorithm and discuess their physical meaning. In Section 4 we validate the algorithm using Monte Carlo radiative transfer simulations. In Section 5 we summarize the results and discuss future work.

2. Expansion Opacity

The total opacity $\kappa_{\rm tot}$ is the sum of contributions from different physical processes:

$$\kappa_{\rm tot} = \kappa_{\rm th} + \kappa_{\rm ff} + \kappa_{\rm bf} + \kappa_{\rm bb} \tag{1}$$

where $\kappa_{\rm th}$ is Thomson scattering, $\kappa_{\rm ff}$ is free-free opacity, $\kappa_{\rm bf}$ is bound-free opacity (which we neglect in this work), and $\kappa_{\rm bb}$ is bound-bound opacity from atomic line transitions.

The BB opacity $\kappa_{\rm bb}$ depends on the plasma's ionization and excitation populations for each material. This is calculated using the Boltzmann distribution.

$$N_{m,i+1} = \frac{N_{m,i}}{N_{m,0}} \frac{Z_{m,i+1}}{Z_{m,i}} \exp\left(-\frac{E_{i+1} - E_i}{kT}\right) \cdot g_e \tag{2}$$

where m, i, j are the indexes of the material, ionization level and energy level respectively, N is the number density, g and E are the degeneracy and energy of the energy level respectively, k is the Boltzmann constant, T is the temperature and g_e is the constant factor $g_e = \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2}$.

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$Z_{m,i}$ is the partition function of the state i for the material m, and is calculated using the partition function formula:

$$Z_{m,i} = \sum_j g_j \exp\left(-\frac{E_j}{kT}\right) \tag{3}$$

The number density $N_{m,i}$ is the calculated using the Saha equation.

$$\frac{N_{m,i}}{N_{m,0}} = \frac{g_i}{g_0}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} \exp\left(-\frac{E_i}{kT}\right) \tag{4}$$

where m_e is the electron mass.

The opactical depth of the bound-bound line is then calculated using the Sobolev approximation.

$$\tau_{j_l,j_u} = t \cdot N_{m,i,j_l} \cdot f_{j_l,j_u} \cdot \frac{\pi e^2}{m_e c} \tag{5}$$

where j_l and j_u are the lower and upper energy levels respectively and f_{j_l,j_u} is the oscillator strength of the transition from the energy level j_l to the energy level j_u, t is the time from explosion.

Then, the opacity is calculated as:

$$\kappa_{BB}(\nu) = \sum_l \delta(\nu - \nu_l) \cdot \tau_l \tag{6}$$

For a steady plasma, the rosseland mean opacity is giving the average distance energy travels before being scattered or absorbed. The rosseland mean opacity is calculated as:

$$\kappa_{\text{Rosseland}} = \frac{\int_0^\infty \kappa_\nu \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu} \tag{7}$$

where B_ν is the Planck function and T is the temperature. The rosseland mean opacity is giving the average distance energy travels before being scattered or absorbed.

The naive application of Rosseland mean opacity to expansion opacity does not work for medium with high BB contribution because baivly calcu

the frequency-dependent opacity changes as the photon propagates through the expanding medium. The distance to the next spectral line is measured in terms of the change in frequency, expressed as a percentage change. Mathematically, the velocity shift is $\Delta v = -c \ln(\nu_0/\nu_{\text{line}})$, where ν_0 is the initial frequency and ν_{line} is the frequency of the next line. This frequency shift corresponds to a velocity shift, and when combined with the time since explosion t, gives the distance $\Delta x = ct \ln(\nu_0/\nu_{\text{line}})$.

In an expanding medium, the path length in frequency space is given by $ds = ctd \ln \nu$, where c is the speed of light, t is the expansion time, and ν is the frequency. For each frequency ν_i, we calculate the distance Δx required to reach optical depth $\tau = 1$:

$$\tau = \int_{x_i}^{x_j} \kappa_\nu \rho dx = 1 \tag{8}$$

where κ_ν is the frequency-dependent opacity and ρ is the density. The smoothed opacity at frequency ν_i is then calculated as:

$$\kappa_{\text{smooth}}(\nu_i) = \frac{1}{\rho \Delta x} \tag{9}$$

where Δx is the path length over which $\tau = 1$ is reached. This approach effectively averages the opacity over a window defined by the expansion physics, capturing the essential behavior of bound-bound line transitions in expanding media.

When using large frequency bins, if the distance required to reach $\tau \geq 1$ is less than a single frequency bin width, the effective mean free path after saturation is calculated according to:

$$\lambda_{\text{eff}} = \frac{1}{\rho \kappa_{\text{eff}}} \tag{10}$$

where $\kappa_{\text{eff}} = 1/(\rho \Delta x)$ and Δx is the path length over which $\tau = 1$ is reached within the bin. This provides a practical way to estimate the effective opacity in each bin when the optical depth saturates before crossing the bin boundary, as illustrated in Figure 1.

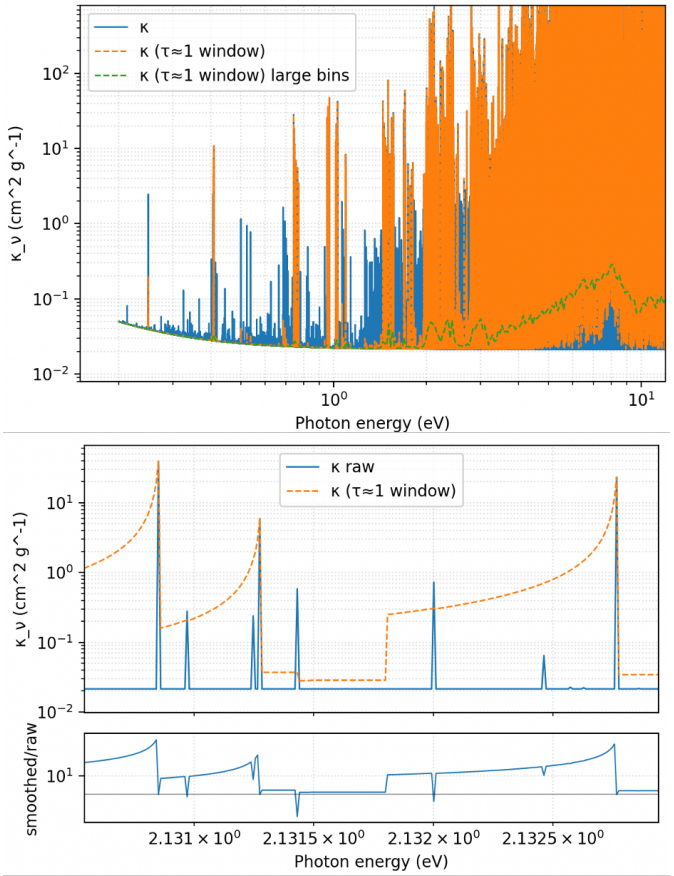


Figure 1: Examples of bound-bound line opacity showing the $\tau = 1$ window smoothing algorithm. Left: illustration of the smoothing process. Right: comparison of raw opacity (solid line), smoothed opacity with fine frequency bins (dashed line), and smoothed opacity with large frequency bins (dotted line), demonstrating the effect of the $\tau = 1$ window smoothing on bound-bound line transitions.

3. Rosseland Mean from Smoothed Opacity

The standard Rosseland mean opacity is defined as:

$$\frac{1}{\kappa_R} = \frac{\int_0^\infty \frac{1}{\kappa_\nu}\frac{\partial B_\nu}{\partial T}\,d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T}\,d\nu} \tag{11}$$

where B_ν is the Planck function and T is the temperature. Using the smoothed opacity $\kappa_{\rm smooth}(\nu)$ from the $\tau = 1$ window algorithm, we compute the expansion-corrected Rosseland mean:

$$\frac{1}{\kappa_{R,\rm exp}} = \frac{\int_0^\infty \frac{1}{\kappa_{\rm smooth}(\nu)}\frac{\partial B_\nu}{\partial T}\,d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T}\,d\nu} \tag{12}$$

4. Validation

Monte Carlo radiative transfer simulations using the new Rosseland mean opacity calculated from the $\tau = 1$ window algorithm yield results identical to full frequency-group calculations. The simulations were performed using URILIGHT [1, 2], an open-source Monte Carlo radiative transfer code designed for supernova ejecta. URILIGHT solves the time-dependent radiative transfer equation in expanding media using a Monte Carlo approach, tracking photon packets through the ejecta and accounting for bound-bound, bound-free, and free-free opacity sources. This demonstrates that the averaging process successfully captures the essential physics of expansion opacity, allowing for efficient gray opacity calculations without loss of accuracy.

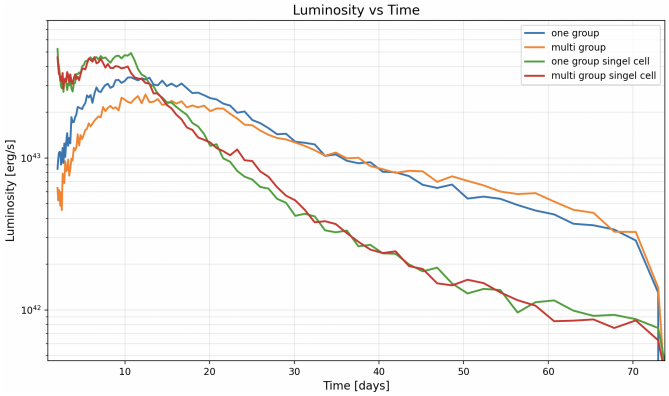


Figure 2: Comparison of Monte Carlo radiative transfer results using the new Rosseland mean opacity (from $\tau = 1$ window algorithm) versus full frequency-group calculations. The agreement demonstrates that the averaging captures the essential physics of expansion opacity.

5. Summary

We have presented a new algorithm for calculating Rosseland mean opacity that accounts for expansion opacity effects from bound-bound atomic transitions. The method uses a $\tau = 1$ window to smooth the frequency-dependent opacity, with the smoothed value calculated as $\kappa = 1/(\rho\Delta x)$ where Δx is the path length to reach optical depth unity. Validation through Monte Carlo simulations confirms that this approach captures the essential physics while enabling efficient gray opacity calculations.

References

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